

# LINEAR INDEPENDENCE OF FOUR TIME–FREQUENCY TRANSLATES AT A RESONANT NON-LATTICE CONFIGURATION

ABSTRACT. Let  $Z$  denote the Zak transform. We prove that if  $g \in L^2(\mathbb{R})$  is nonzero and  $Zg$  has a continuous representative — in particular if  $g$  is a nonzero Schwartz function — then the four translates  $g(x)$ ,  $g(x-1)$ ,  $e^{2\pi i x}g(x)$ ,  $e^{2\pi i \sqrt{2}x}g(x-\sqrt{2})$  are linearly independent. This settles the case  $\Lambda_0 = \{(0,0), (1,0), (0,1), (\sqrt{2}, \sqrt{2})\}$  of the Heil–Ramanathan–Topiwala conjecture, listed as Conjecture 2 in the survey of Heil and Speegle. More generally we prove independence for  $\{(0,0), (1,0), (0,1), (\alpha, \alpha+j)\}$  for every irrational  $\alpha$  and every integer  $j \notin \{-1, 1\}$ , and we show that  $j = \pm 1$  is precisely where the method saturates.

The configuration  $\Lambda_0$  is *resonant*:  $-\sqrt{2} + \sqrt{2} = 0$ , so the relevant translation of  $\mathbb{T}^2$  is not ergodic and  $\mathbb{T}^2$  fibres into invariant circles. This fibration, the resulting dichotomy, and the value of the fibre winding number are due to Oussa [10]; see the Attribution section for a precise account. Our contribution is the endgame: we read the winding value as a root count, and then extract from Jensen’s formula a quantity that is *constant across fibres* and contradict that constancy by Vieta’s relations.

## 1. INTRODUCTION

The HRT conjecture asserts that for every nonzero  $f \in L^2(\mathbb{R})$  and every finite set of distinct points  $\{(a_k, b_k)\} \subset \mathbb{R}^2$  the functions  $e^{2\pi i b_k x} f(x - a_k)$  are linearly independent. It is known for at most three points [4], for subsets of a lattice [7], for configurations lying on two parallel lines [2, 3], for almost every  $(1, 3)$  configuration [8], and for various restricted window classes [1, 9]. The smallest configuration resisting all of these is

$$\Lambda_0 = \{(0,0), (1,0), (0,1), (\sqrt{2}, \sqrt{2})\},$$

singled out as Conjecture 2 (“the HRT subconjecture”) in the survey of Heil and Speegle [5], and open for Schwartz windows. Our main result settles it.

**Theorem 1.** *Let  $g \in L^2(\mathbb{R})$ ,  $g \neq 0$ , and suppose the quasi-periodic extension of  $Zg$  given by (1) is continuous on all of  $\mathbb{R}^2$ . Then the four functions  $g(x)$ ,  $g(x-1)$ ,  $e^{2\pi i x}g(x)$ ,  $e^{2\pi i \sqrt{2}x}g(x-\sqrt{2})$  are linearly independent. In particular this holds for every nonzero Schwartz  $g$ , and more generally for  $g$  in the Wiener amalgam  $W(C, \ell^1)$ .*

**Theorem 2.** *Let  $\alpha$  be irrational and  $j \in \mathbb{Z}$  with  $j \neq \pm 1$ . Then for every nonzero  $g$  with continuous Zak transform, the translates indexed by  $\{(0,0), (1,0), (0,1), (\alpha, \alpha+j)\}$  are linearly independent.*

*Remark 3.*  $\Lambda_0$  is not reachable from the solved cases. Affine symplectic maps preserve HRT dependences, and the difference vectors of  $\Lambda_0$  have symplectic invariants  $1, \sqrt{2}, -\sqrt{2}$ ; membership in a lattice would force these into a common  $d\mathbb{Z}$ , impossible since  $1/\sqrt{2}$  is irrational. Parallelism and collinearity are affine invariants and fail for  $\Lambda_0$ . Hence Linnell's theorem, the two-parallel-line results, and the collinear results cannot be applied after any change of coordinates.

## 2. REDUCTION

Normalise  $Zg(t, \omega) = \sum_{n \in \mathbb{Z}} g(t - n)e^{2\pi i n \omega}$ , so that

$$(1) \quad Zg(t + 1, \omega) = e^{2\pi i \omega} Zg(t, \omega), \quad Zg(t, \omega + 1) = Zg(t, \omega),$$

$Z$  is unitary onto  $L^2(\mathbb{T}^2)$ , and  $Z(M_b T_a g)(t, \omega) = e^{2\pi i b t} Zg(t - a, \omega - b)$ .

Assume a nontrivial dependence with coefficients  $c_1, \dots, c_4$  for the configuration  $\{(0, 0), (1, 0), (0, 1), (\alpha, \alpha + j)\}$ . If some  $c_i = 0$  the dependence is supported on at most three points; for one or two points this is immediate, for three collinear points it is the theorem of Heil–Ramanathan–Tapiwala, and for three points in general position an element of  $SL(2, \mathbb{R})$  carries the difference vectors to  $(1, 0)$  and  $(0, s)$  with  $s = \omega(u_2, u_3) \neq 0$ , placing the configuration inside the lattice  $\mathbb{Z} \times s\mathbb{Z}$ , where Linnell's theorem applies. Metaplectic operators are unitary on  $L^2$ , preserve the Schwartz class, and carry time–frequency translates to time–frequency translates up to unimodular factors, so dependences correspond. We may therefore assume all  $c_i \neq 0$ .

Writing  $G = Zg$  and using (1) together with  $j \in \mathbb{Z}$ , the dependence becomes

$$(2) \quad p(t, \omega)G(t, \omega) = -c_4 e^{2\pi i(\alpha+j)t} G(t - \alpha, \omega - \alpha), \quad p = c_1 + c_2 e^{-2\pi i \omega} + c_3 e^{2\pi i t}.$$

Note that  $j$  has disappeared from the shift and survives only in the prefactor.

The translation  $\gamma = (-\alpha, -\alpha)$  of  $\mathbb{T}^2$  is *not* ergodic:  $e^{2\pi i(\omega-t)}$  is a nonconstant invariant character. Its ergodic components are the circles  $L_c = \{\omega = t + c\}$ , on which  $\gamma$  acts as rotation by  $\alpha$ ; the parameter  $c = \omega - t \bmod 1$  is well defined on  $\mathbb{T}^2$ , and each  $L_c$  is a single circle. Set  $G_c(t) = G(t, t + c)$  and  $z = e^{2\pi i t}$ . Then (1) and (2) restrict to

$$(3) \quad G_c(t + 1) = e^{2\pi i(t+c)} G_c(t),$$

$$(4) \quad G_c(t - \alpha) = -c_4^{-1} e^{-2\pi i(\alpha+j)t} P_c(t) G_c(t), \quad P_c(t) = z^{-1} Q_c(z),$$

where

$$(5) \quad Q_c(z) = c_3 z^2 + c_1 z + c_2 e^{-2\pi i c}.$$

Only the phase of the constant term of  $Q_c$  depends on  $c$ ; this drives everything below.

### 3. THE FIBRE DICHOTOMY

**Lemma 4.** *Each fibre satisfies either  $G_c \equiv 0$  or  $G_c(t) \neq 0$  for all  $t$ . The set  $C_1 = \{c : G_c \text{ nowhere zero}\}$  is open, and if it is empty then  $g = 0$ . Consequently, for a nonzero  $g$ ,  $C_1$  contains an interval, and  $P_c$  is zero free on the circle for every  $c \in C_1$ .*

*Proof.* By (4),  $G_c(t_0) = 0$  forces  $G_c(t_0 - \alpha) = 0$ , hence  $G_c(t_0 - n\alpha) = 0$  for all  $n \geq 0$ ; no division occurs. Since  $\alpha$  is irrational this orbit is dense modulo 1, and  $|G_c|$  is 1-periodic by (3), so continuity gives  $G_c \equiv 0$ . By (1) the modulus  $|G|$  is a well-defined continuous function on  $\mathbb{T}^2$ , so  $\{|G| > 0\}$  is open;  $C_1$  is its image under the bundle projection  $\mathbb{T}^2 \rightarrow \mathbb{T}$ ,  $(t, \omega) \mapsto \omega - t$ , which is an open map. If  $C_1 = \emptyset$  then  $G \equiv 0$ , so  $g = 0$ . A nonempty open subset of  $\mathbb{T}$  contains an interval. The last claim follows from (4), since  $G_c$  is nowhere zero.  $\square$

### 4. THE DEGREE IDENTITY

**Lemma 5.** *For every  $c \in C_1$  the polynomial  $Q_c$  has exactly  $j + 1$  roots in the open unit disc. In particular  $C_1 = \emptyset$ , and hence  $g = 0$ , whenever  $j \notin \{-1, 0, 1\}$ .*

*Proof.* Let  $m_c(t) = e^{2\pi i c t} e^{\pi i t(t-1)}$ , so that  $m_c(t+1)/m_c(t) = e^{2\pi i(t+c)}$  and  $v_c = G_c/m_c$  is 1-periodic by (3). From  $t(t-1) - (t-\alpha)(t-\alpha-1) = 2\alpha t - \alpha^2 - \alpha$  we get  $m_c(t)/m_c(t-\alpha) = e^{2\pi i c \alpha} e^{2\pi i \alpha t} e^{-\pi i(\alpha^2 + \alpha)}$ , and the non-periodic exponentials cancel:  $e^{-2\pi i(\alpha+j)t} \cdot e^{2\pi i \alpha t} = e^{-2\pi i j t}$ . Hence (4) becomes

$$(6) \quad v_c(t-\alpha) = R_c(t)v_c(t), \quad R_c(t) = \Lambda_c e^{-2\pi i j t} P_c(t),$$

with  $\Lambda_c$  a nonzero constant and  $R_c$  continuous, 1-periodic and zero free.

Put  $w_c = v_c/|v_c|$  and  $\eta_c = R_c/|R_c|$ , continuous maps  $\mathbb{T} \rightarrow \mathbb{T}$ , so  $w_c(t-\alpha) = \eta_c(t)w_c(t)$ . Taking degrees, and using that precomposition with a rotation does not change the degree of a circle map, gives  $\deg w_c = \deg \eta_c + \deg w_c$ , whence

$$(7) \quad \deg \eta_c = 0.$$

Since  $\deg \eta_c = -j + \deg(P_c/|P_c|)$  and, by the argument principle applied to  $P_c = z^{-1}Q_c$  on the zero-free circle,  $\deg(P_c/|P_c|) = -1 + N(c)$  with  $N(c)$  the number of roots of  $Q_c$  in the disc, we get  $N(c) = j + 1$ . As  $Q_c$  is quadratic,  $N(c) \in \{0, 1, 2\}$ , which is impossible for  $j \notin \{-1, 0, 1\}$ .  $\square$

### 5. CROSS-FIBRE CONSTANCY AND THE CONTRADICTION

From here take  $j = 0$ , so  $N(c) = 1$  for all  $c \in C_1$ :  $Q_c$  has one root  $\zeta_{\text{in}}(c)$  inside and one root  $\zeta_{\text{out}}(c)$  outside the unit circle, and both depend continuously on  $c$  on  $C_1$  because the two roots stay separated.

**Lemma 6.**  $|\zeta_{\text{out}}(c)| = |c_4|/|c_3|$  for every  $c \in C_1$ .

*Proof.* On a fibre in  $C_1$  the function  $\log |G_c|$  is continuous and, by (3), 1-periodic. Taking  $\log |\cdot|$  in (4) and integrating over a period, the  $G$  terms cancel by translation invariance of Lebesgue measure alone, leaving  $\int_0^1 \log |P_c| = \log |c_4|$ . Since  $\int_0^1 \log |z^{-1}| = 0$ , Jensen's formula gives  $\int_0^1 \log |P_c| = \log |c_3| + \log |\zeta_{\text{out}}(c)|$ .  $\square$

*Proof of Theorems 1 and 2.* Theorem 2 for  $j \neq \pm 1$  is Lemma 5. For  $j = 0$  suppose a nontrivial dependence exists; by Lemma 4 we may work on an interval  $I \subseteq C_1$ . Write  $r = |c_4|/|c_3|$  and  $S = -c_1/c_3$ . By Vieta,  $\zeta_{\text{in}} + \zeta_{\text{out}} = S$  is constant on  $I$  and  $\zeta_{\text{in}}\zeta_{\text{out}} = (c_2/c_3)e^{-2\pi ic}$ . By Lemma 6,  $|\zeta_{\text{out}}| \equiv r$ , hence also  $|\zeta_{\text{in}}| = |c_2|/(|c_3|r)$  is constant, and therefore  $|\zeta_{\text{out}} - S| = |\zeta_{\text{in}}|$  is a positive constant  $r'$  (positive because  $\zeta_{\text{in}}\zeta_{\text{out}} \neq 0$ , as  $c_2 \neq 0$ ). Thus  $\zeta_{\text{out}}(c)$  lies in

$$\{|z| = r\} \cap \{|z - S| = r'\},$$

the intersection of two circles with distinct centres, since  $S \neq 0$  as  $c_1 \neq 0$ . This set has at most two points. A continuous function on the interval  $I$  with values in a finite set is constant, so  $\zeta_{\text{out}}$ , and hence  $\zeta_{\text{in}}$ , is constant on  $I$ ; therefore the product  $(c_2/c_3)e^{-2\pi ic}$  is constant on  $I$ , which is false since  $c_2 \neq 0$  and  $I$  has positive length. Theorem 1 is the case  $\alpha = \sqrt{2}$ ,  $j = 0$ , after the shift  $(\sqrt{2}, \sqrt{2})$ .  $\square$

## 6. WHY $j = \pm 1$ SATURATES THE METHOD

For  $j = \pm 1$  Lemma 5 gives  $N(c) \in \{0, 2\}$ : both roots lie on the same side of the circle. Jensen's formula then returns  $\log |c_3|$  (if  $N = 2$ ) or  $\log |c_2|$  (if  $N = 0$ ), so Lemma 6 degenerates to the single  $c$ -independent condition  $|c_3| = |c_4|$ , respectively  $|c_2| = |c_4|$ , and the rigidity argument has nothing to act on.

Two further routes are unavailable. First, the degree-based obstructions for cocycles over irrational rotations (Anzai; Iwanik–Lemańczyk–Rudolph) require a cocycle of nonzero degree, whereas (7) pins  $\deg \eta_c = 0$  for *every*  $j$ : the degree is forced by the mere existence of a continuous 1-periodic solution. (For merely measurable windows there is no degree, nothing forces 0, and these theorems do apply — which is how the  $L^2$  statement is reached by a different argument.) Second, no single fibre obstructs: for  $N = 2$  one has  $R_c = \Lambda_c c_3(1 - \zeta_1/z)(1 - \zeta_2/z)$ , so  $\log R_c$  has Fourier support in nonpositive frequencies with geometrically decaying coefficients, and the cohomological equations  $\hat{\theta}(n)(e^{2\pi i n \alpha} - 1) = \hat{\psi}(n)$  are solvable for all  $n \neq 0$ , the mean condition  $\hat{\psi}(0) = 0$  being exactly Lemma 6. For badly approximable  $\alpha$  the small divisors are dominated by the geometric decay. Root location is likewise unobstructed: with  $\zeta_1 + \zeta_2 = S$  fixed and  $|\zeta_1\zeta_2|$  constant, the Schur–Cohn conditions are satisfiable on a short interval.

Consequently a proof for  $j = \pm 1$  must use global input that the above never needs: the quasi-periodicity (1) makes  $G$  a section of a line bundle over  $\mathbb{T}^2$  of Chern number one, so every continuous Zak transform vanishes

somewhere and dead fibres necessarily exist; and the fibrewise solutions must be glued into a single continuous  $G$  with matching behaviour on  $\partial C_1$ .

#### ATTRIBUTION

This paper builds directly on the program of Oussa [10], and we want the division of credit to be unambiguous. Already present there, for this configuration family and for Schwartz windows: the Zak-transform reduction (2); the non-ergodicity of  $\gamma$  and the resulting fibration of  $\mathbb{T}^2$  into invariant circles; the zero-propagation dichotomy of Lemma 4; and — this is the point we wish to stress — *the value of the fibre winding number*. Specifically, the proof of Proposition 6.2 of [10] derives an identity which, specialised to  $d = 1$ , fourth point  $(\alpha, \alpha + j)$ ,  $a = (-1, 1)$  and  $m = 1$ , yields exactly  $\kappa_a = j$ ; the winding lemma there is in turn credited to suggestions of C. Heil, K. Okoudjou and R. Jadhav. Oussa states the integrality conclusion (remarking that it is often automatic in dimension one) and draws no root-count consequence, and he names  $\Lambda_0$  as the unresolved resonant case.

Accordingly we claim only the following. (i) A derivation of (7) using *continuity alone*, with no smoothness hypothesis, via the observation that the degree of a circle map is unchanged by precomposition with a rotation. (ii) The reading of that identity as the *root count*  $N(c) = j + 1$ , which immediately gives Theorem 2 for  $|j| \geq 2$  by a parity-of-degree count. (iii) The evaluation of the fibre mean by Jensen’s formula so as to produce a quantity that is *constant across fibres* (Lemma 6). (iv) The Vieta rigidity contradicting that constancy, which is what actually settles  $\Lambda_0$ . Items (iii) and (iv) are, to our knowledge, new; neither Jensen’s formula nor Vieta’s relations appear in [10].

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